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Chapter 1
Change Management, Organizational Adaptation, and Labor Market Restructuration:
Notes for the Post-COVID-19 Era

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ABSTRACT

The recent transformation caused by the COVID-19 pandemic crisis drives the world economy to an accelerated mutation. This chapter focuses on how the current developments affect the various socio-economic organizations and systems and how they can adapt to this new emerging reality. To this end, relevant forecasts on the current pandemic crisis are examined. This crisis seems to cause the acceleration of the Fourth Industrial Revolution, functioning as a catalyst of the structural changes also observed in the working environment. The chapter suggests that all socioeconomic organizations (irrespectively of their size, spatial reach, and sectorial focus) are called upon nowadays to readjust themselves and that innovation is the fundamental generator for exiting the ongoing structural crisis. However, innovation unavoidably creates significant changes that socioeconomic organizations must manage effectively in the foreseeable future, according to a new way of perceiving organizational resilience and adaptability for the post-COVID-19 era.

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INTRODUCTION

The COVID-19 (hereafter pandemic) crisis leads to a stimulation of developments, facilitating the emergence of a repositioned framework for all levels of today's global economy (Soto-Acosta, 2020; Syani et al., 2020). Within the same, globally transpositioned context, the Fourth Industrial Revolution (4IR) increasingly becomes a force for disruption for various older and relatively saturated—in broad socioeconomic and innovational terms—professions and industries (Bonilla-Molina, 2020; Tiwari et al., 2021). In this profound transformation and paradigm shift for all socioeconomic organizations on the planet, adaptation, survival, and growth can only be uncertain and demanding (Marinov & Marinova, 2021; World Bank, 2020b). According to these general considerations on the global system's evolution, the question arising is how the socioeconomic organizations affected by the current trends can adapt to this emerging global mutation.

To explore this question and propose solutions, studying the unfolding global pandemic crisis seems to be imperative. This conceptual article will attempt an elliptic examination of the related literature, presenting projections by international organizations on the pandemic and significant developments in labor relations, aiming to offer new theoretical insight on the subject of organizational adaptation (Jaakkola, 2020; Snyder, 2019). In section two, scenarios and forecasts by international organizations are examined, oriented towards the global socioeconomic system's immediate future amid the pandemic crisis. In section three, the basic features of the 4IR are emphasized. The aim is to reaffirm whether the 4IR is accelerated based on the structural changes observed in labor relations. This is the context that leads us to conclude that organizational adaptation (OA) is the predominant ingredient required for the future development of all socioeconomic organizations affected by the crisis. Finally, in section five, conclusions are drawn, suggesting directions for future research.

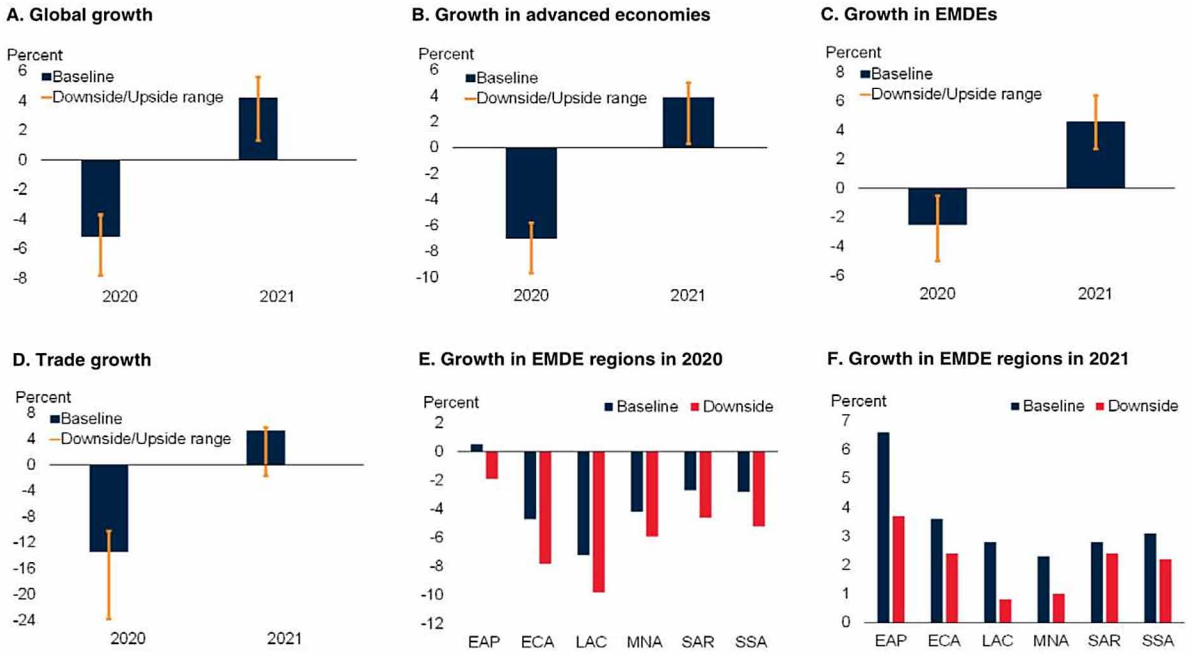
EMPIRICAL BACKGROUND OF THE COVID-19 CRISIS

Following the worldwide spread of the pandemic since the early days of 2020, the global socioeconomic system has entered a period of unprecedented turbulence and regression. According to António Guterres, Secretary-General of the United Nations (2020), this pandemic crisis leads the whole world to a readjustment. Humanity is in front of extraordinary developments, where the weakest social groups and communities increasingly become more vulnerable. Guterres also speaks of the required swift action for our world to avoid the spread of other diseases and protect from violent extremists to rise in power.[1]

According to the World Bank (2020a), which predicted a slight rebound for the global economy in 2021 based on a series of forecasts, the pandemic crisis has triggered a decisive blow to the already fragile world economy, with the magnitude of the recession being the largest since World War II. More specifically, this report by the World Bank foresaw three scenarios of possible economic recovery for 2020-2021: one consistent with baseline forecasts, a downside of a protracted global recession, and an upside that projects a prompt recovery (Figure 1).

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Figure 1. Forecasted global growth outcomes as reproduced from World Bank (2020a) for emerging markets and developing economies (EMDE), East Asia and Pacific (EAP), Europe and Central Asia (ECA), Latin America and the Caribbean (LAC), Middle East and North Africa (MNA), South Asia (SAR), and Sub-Saharan Africa (SSA).

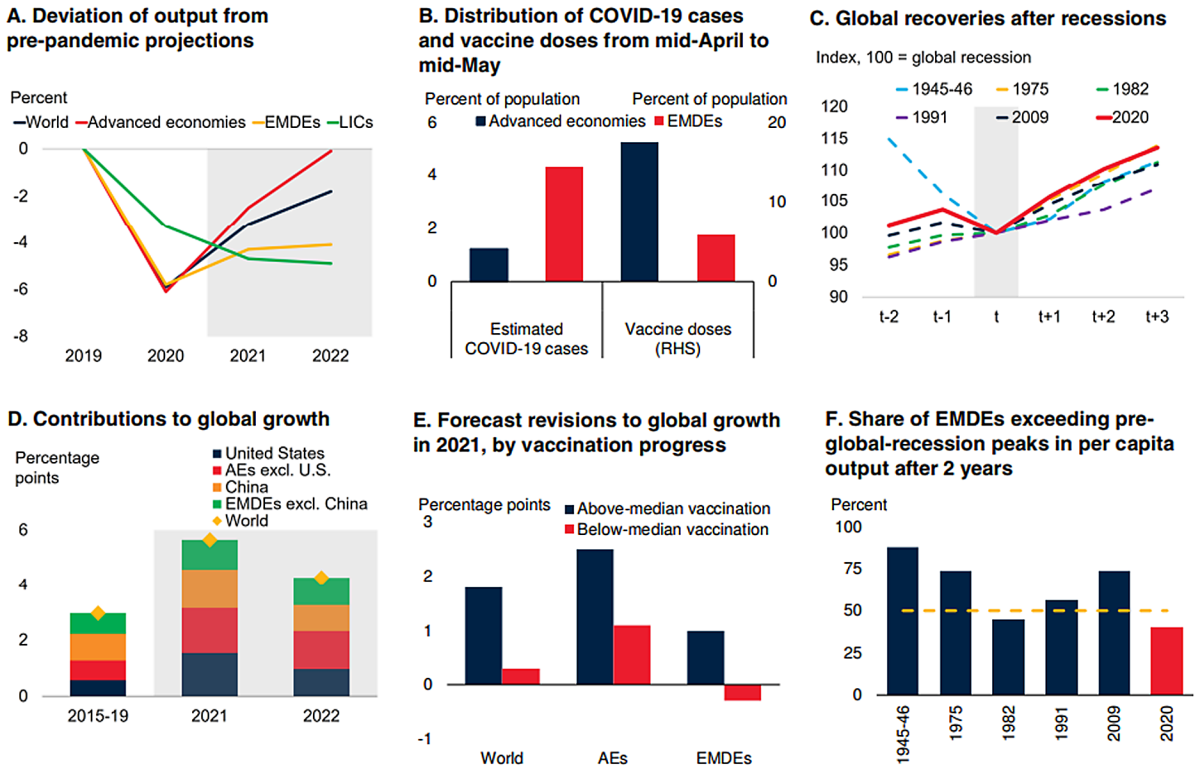


The baseline scenario predicted that global output would contract by 5.2 percent and international trade by 13 percent in 2020 due to the unprecedented disruption of global value chains, the downside scenario that the prolongation of stricter lockdown measures would result in the fall of global output in 2020 by 8 percent, and the upside scenario of a prompt recovery following the quarantine and most health protection measures. The study concluded that the health crisis would leave its mark on investor and consumer confidence; therefore, policymakers need—in addition to continuing with short-term measures to tackle the pandemic—to design comprehensive programs to stimulate and improve the fundamental drivers of economic growth.

Finally, the global economy moved in the upside scenario's range since, as noticed by the updated respective report by the World Bank (2021), the global output's contraction was 3.5 percent. The advanced economies recorded a 4.7 percent recession, the emerging market and developing economies 1.7 percent, respectively, and international trade fell by 8.3 percent. In aggregate, the updated report projects an uneven rebound for the various economies and socioeconomic actors, especially for impoverished countries (for example, Pattenden et al., 2021, notice that the pandemic's impact mainly worsened the conditions for the labor classes involved in precarious casual labor, reproductive labor, and petty commodity production). This uneven recovery is (to some extent) mitigated by the projected significant global growth of 5.6 percent in 2021. However, the report forecasts that this scenario (by also considering the significant projected global growth of 2022) will not drive the global economy to surpass the pre-pandemic expectations and, as a result, the current global recession does not lead to a usual recovery cycle since growth inequalities are unprecedentedly intensified (Figure 2).

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Figure 2. Forecasted global economic prospects for the world, the advanced economies (AEs), the emerging markets and developing economies (EMDE), and the low-income countries (LICs).
Reproduced and minor edited (only changed the figures' order of appearance) from the World Bank (2021).

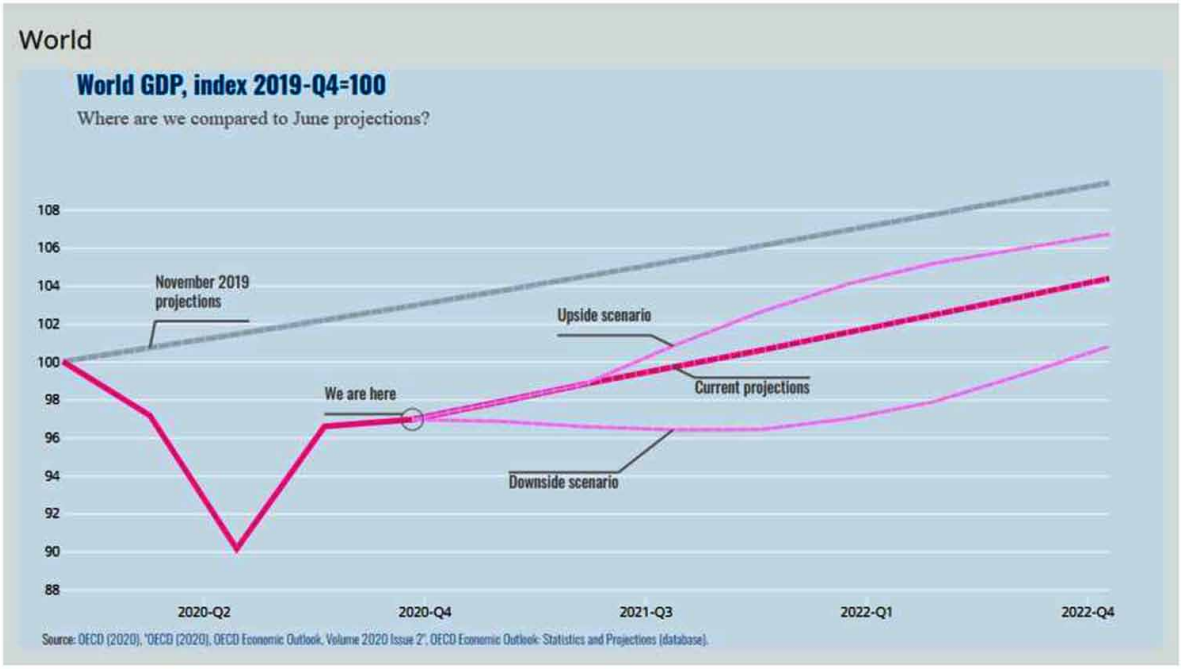


In a similar vein, the International Monetary Fund (IMF, 2020) described this crisis as "like no other," having forecasted for 2021 a slight rebound based on specific scenarios. The International Labor Organization (ILO, 2020) pointed out that working-hour losses due to quarantine measures and social distancing have accounted for four hundred million full-time jobs. Accordingly, the World Trade Organization (Azevêdo, 2020) has forecasted that the COVID- 19 recession and socioeconomic crisis is expected to surpass the preceding economic crisis of 2008-2009 in most indices. A spread and reinforcement of underdevelopment is another concern of COVID-19 dynamics, with specific international organizations noticing that new forms of exclusion and lagging might occur in the foreseeable future (FAO, 2020; OECD, 2020b). Also, various industries and economic sectors, mostly in developing countries, seem to be violently readjusted, entering into a trajectory of profound transformations due to a loss of significant value-added (International Civil Aviation Organization, 2020; International Energy Agency, 2020; Matthewman & Huppatz, 2020; WHO Director-General, 2020).

In this reframed global context, an issue that actively concerns scholars and policymakers is the shape the recovery of the world economy will take in the future. For example, the OECD (2020a) suggested a V-shaped return for global GDP in 2021 and 2022 after a steep decline in 2020 (Figure 3).

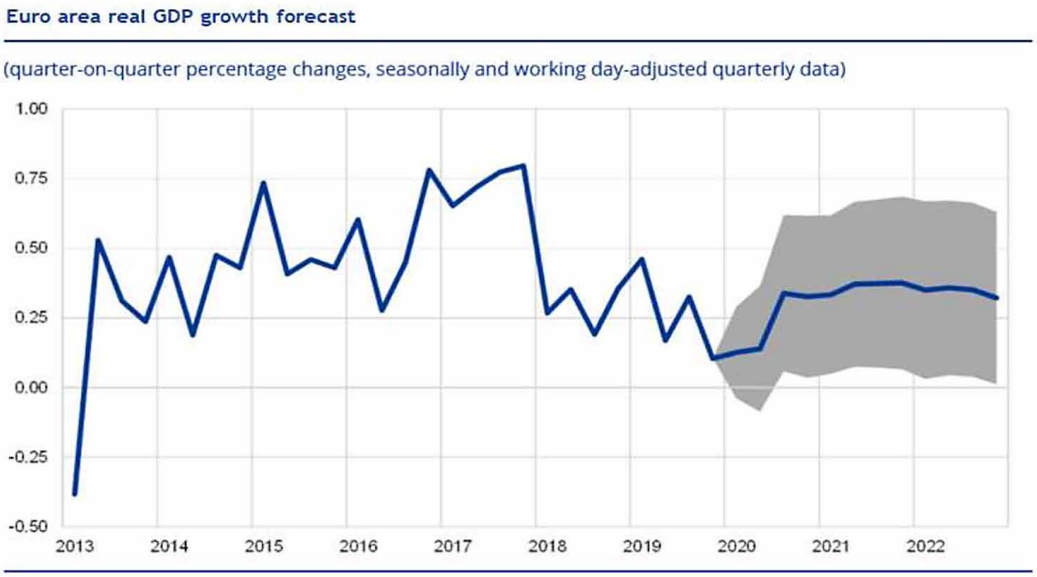
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Figure 3. OECD V-shaped forecast.
Source: OECD (2020a) and OECD Economic Outlook: Statistics and projections database.



Also, the European Central Bank (ECB, 2020) forecasted a more moderated version of recovery for the euro area, pointing out that it is impossible to evaluate the full impact of the pandemic. According to this forecast, real GDP growth in the euro area is not expected to increase more than 1.3 percent in 2021 and 1.4 percent in 2022 (Figure 4).

Figure 4. Euro area real GDP growth forecast.
Source: ECB (2020).



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Although various scholars appear relatively recently to support a V-shaped rebound for the world economy, which suggests a rapid recovery after the recession, this seems increasingly infeasible (Beech, 2020; Gómez-Pineda, 2020; Gregory et al., 2020; Lawrence & Homer-Dixon, 2020). Unfortunately, an L-shaped rebound seems the likely outcome for various cases since the less-developed socioeconomic systems that lack resilience, adaptability, and innovation might not benefit when international trade begins to recover in the restructured global and sectoral environment (Arbolino & Caro, 2021). Overall, it seems that the world has entered an irreversible trajectory of rebalances, with the post-pandemic recovery appearing unequal across the different regions and socioeconomic actors around the globe.

THE WORLD OF WORK AND PRODUCTION IS CHANGING PROFOUNDLY AMID THE FOURTH INDUSTRIAL REVOLUTION

A well-known approach to studying these radical global reversals is the Fourth Industrial Revolution (4IR).[2] Industry 4.0 was the first related term, and it was suggested in a high-tech strategy project by the German government to support manufacturing's computerization (BMBF-Internetredaktion, 2016). The subsequent invention of the "Fourth Industrial Revolution" concept is primarily attributed to K. Schwab, Founder and Executive Chairman of the World Economic Forum, who used the term in an article related to foreign affairs and later at a meeting in Davos in 2016. Schwab (2015) introduced this concept by suggesting that our world is in front of a technological revolution that will profoundly change how we live, work, and relate with each other. This transition will not resemble anything humanity has encountered so far concerning the scale, scope, and complexity. In the same analysis, Schwab also noticed that the response to our world's shared issues must be integrated and comprehensive, including all private and public stakeholders and from all fields. The 4IR follows the sequence of earlier developments in human history, oriented primarily at worldwide technology application and diffusion in paradigmatic terms (Perez, 2010). The first industrial revolution used water and steam power to mechanize production. Electric power was used by the second to produce mass output, while the third used electronics and information technology to automate manufacturing. Nowadays, the 4IR builds upon the third, driving forward the digital revolution that started in the late twentieth century (Braña, 2019; Loureiro, 2018).

The most extraordinary element noticed by 4IR scholars is the diffusion and evolution of the "cyberphysical systems," meaning the fusion of technologies that drives an increasing blurring between the boundaries of the physical, digital, and biological spheres (Gill, 2008; Schlick, 2012). Schwab (2016) also notices that humanity is faced with the opportunity of income increase and diffusion of cost-efficient products and services, accompanied by the threat of manual labor substitution by automation and exacerbation of social tensions and inequality. Concerning the impact in business, Schwab argues that it becomes progressively challenging to predict what changes innovation brings forward and, therefore, incumbents will be disrupted at a faster pace than before by the newcomers. This development reinforces the demand-side drastically, giving consumers new options.[3]

By reflecting on this changing context of the global socioeconomic system, the emerging challenges in labor relations are presented below. Such an analysis would allow us to discern the acceleration of the 4IR due to the global pandemic crisis. Therefore, concerning the labor market's ongoing reorganization, according to the ILO (2021d), the numbers of teleworking—that used to be in the low single digits—have grown immensely after the COVID-19 outbreak. Post-COVID-19 teleworking accounts for more than 33 percent in Europe, approaching 50 percent in the United States. In a related report about the impact

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of COVID-19 on wages, the ILO (2021b) has estimated a 10.7 percent decline in global labor income for the first three quarters of 2020 compared to the same period of 2019. In the same report, the ILO concludes that the COVID-19 pandemic crisis did not affect all industries the same, with the sectors hardest hit being wholesale and retail trade, accommodation and food services, and other sectors where women tend to be the working majority.

Regarding more specific issues in the working sphere, the pandemic has caused poor human relations due to a lack of meetings and interactions among peers within enterprises. Simultaneously, it may deliberate illegal practices to avoid changes in regulating employment relations and hiring bans (Ostrovskaia, 2020). Many firms have tried to transform the legal status of employment relationships to a business-to-business (B2B) relation between a company and an independent contractor in those turbulent circumstances because this B2B practice helps them reduce taxes, regulations, and employers' commitments to the workers (Mora Cortez & Johnston, 2020).

Nevertheless, the parties concerned have declared a common will to continue a more cooperative relation during the post-COVID-19 period. Thus, a vital recommendation by the ILO (2021a) is that policymakers should implement recovery strategies based on social dialogue procedures, promoting the transition to a more inclusive, resilient, and sustainable world of work. Furthermore, tripartite concertations are expected to face considerable challenges in the new labor market landscape since the venture of working from home during the last months of the pandemic has become the new normal for many bread-earners and trade unions appear less adaptive to this new environment than employers (ILO, 2021c).

In this post-COVID-19 context, employers should accept the new scenery by dealing with employees' work-related stress, managing health and safety beyond the workplace, monitoring performance, reengineering working schedules, and coordinating personnel remotely. Moreover, workers that choose to work from a distance in a lower-cost zone could be paid less than their colleagues commuting from higher-cost areas. Those jobs that are not required to be located in expensive metropolitan work environments could move to low-cost in-country workplaces or beyond the country borders, i.e., service off-shoring. Therefore, the degree of that kind of "telemigration" depends on the "teleworkability" of each occupation itself (Sostero et al., 2020). Furthermore, remote employees have repeatedly stressed that they feel they do not do a constructive job and are isolated. This trend is exacerbated considering the difficulties of separating working and non-working life, especially when work and domestic tasks and responsibilities co-exist in the same place. Stress, psychosocial risk, and uncertainty are significant issues that multiply their dynamic nowadays since working outside the regular working environment can lead to longer, unsocial, and irregular working schedules (Eurofound, 2020).

Another trend that must be considered is that national labor legislation and industrial relations institutions have been developed around worksites. Therefore, nowadays, trade unions should avoid dispersing workers since this can weaken their rights and bargaining power towards the employer-side (Markey, 2020). In this context of profound alterations in the labor market, concerns have been raised that the home-based workers will undertake a part of the costs (i.e., cost of equipment, broadband availability). For the rest of the workers, an improvement of occupational safety and health measures, including organizational changes, avoiding crowding, and providing protective equipment, is needed (Schall & Chen, 2021). Concerning the organization-wide changes, these refer to work practices that include working from home, virtual teamwork, leadership, and management. This trend has been accelerated by the urgent necessity for social distancing provisions, which has obliged various companies to decentralize their functions in regional business units. This digital transformation is a challenge for employee commit-

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ment and engagement since poor communication or a lack of face-to-face interactions can cause further deterioration of employee relations (Collings et al., 2021).

The post-pandemic working environment has expanded the opportunity of remote work among younger and lower qualified employees primarily, and these occupational gaps (digital divide) have been reflected in remarkable differences regarding wage and educational level in teleworkability. This kind of development could foster the division between a "digital precariat" and a "labor aristocracy," and, thus, if the latter separates into an upper sphere of the labor force, social distances might be much more extensive, and social integration could be less successful (Calderón-Gómez et al., 2020; Sutherland et al., 2020). Finally, it should also be mentioned that recent studies have shown that older employees are more likely to comply with preventive measures against COVID-19 and advance their knowledge and technical skills if they need to work from home, which means that they can be better prepared for the digital competencies that will be necessary at the workplaces in the future (Kooij, 2020).

These developments lead us to conclude that the 4IR is accelerating today. Mixed forms of face-to-face and remote working and organizational readjustment have become a reality for many socioeconomic organizations. The bottom line is that many workers may find themselves at lower incomes, which will lead to an increase in inequalities. This adverse development results from shortcomings (in employability terms) in new knowledge and skills to meet the intellectual capital requirements in modern organizations (Andrikopoulos, 2010). These are manifested in vulnerable workers employed in less-adaptable local business ecosystems and lower-competitiveness industries. What remains to be explained is how we can tackle this crisis and gain improved returns (as employees, as companies, and, more generally, as socioeconomic organizations). The following section focuses on intra-business and intra-organizational issues of a strategic nature, whose reevaluation for the post-COVID-19 era seems necessary.

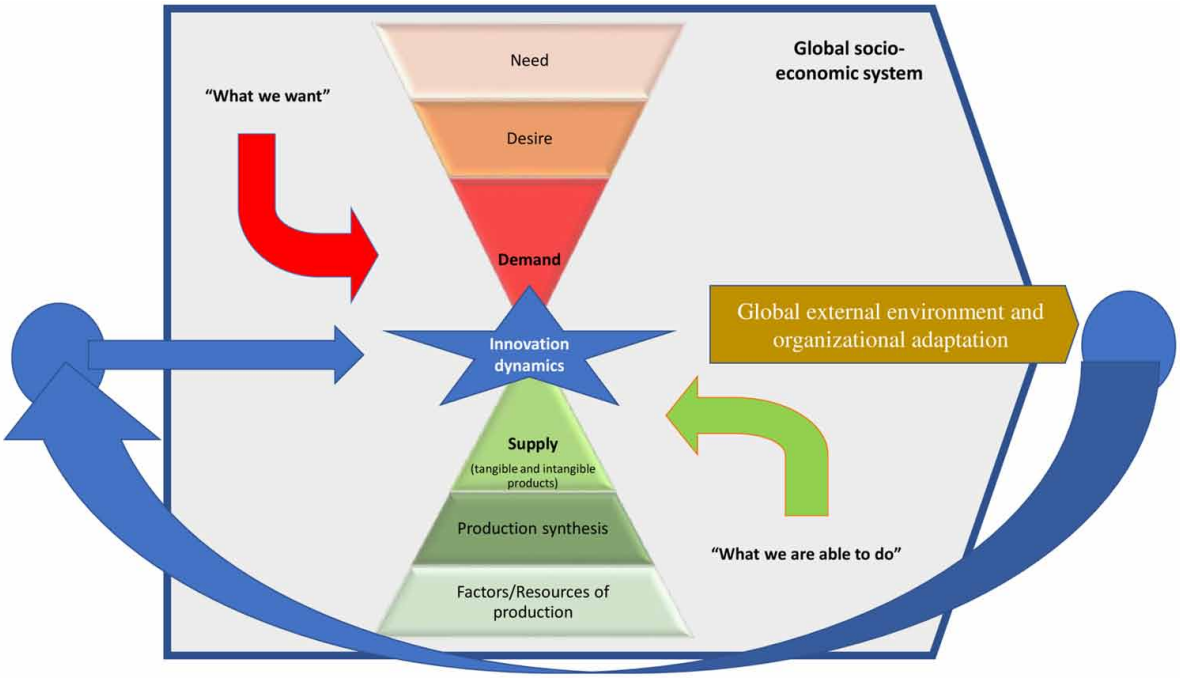
HOW CAN THE AFFECTED SOCIOECONOMIC ORGANIZATIONS READAPT?

By delving into the firms' micro dimensions, it can be argued that the pandemic has encouraged flexible working arrangements through rapid consultations. Current working conditions have offered new opportunities for employee voice and labor-management partnership schemes for designing and implementing effective anti-crisis responses (Merone & Whitehead, 2021). This acceleration of organizational innovations—or deteriorations—would not have reached this scale if the pandemic crisis had not outbroken and spread across the world. What should now concern employees and employers (and any other socioeconomic organization) is generating innovation and improving performance, capable of leading out of any vicious circle spiral. However, innovation always creates change, which must be managed effectively to bring positive results for the whole organization.

Therefore, all organizational changes must be examined under the prism of the required innovation in generic terms (diagonal, vertical, and horizontal; Torfing et al., 2012). From a neo-Schumpeterian perspective, innovation is the "creative destruction" of older and aging organizational configurations due to the "entrepreneur's ambitions" who, in search of profit, is driven to the introduction of new products and new production and marketing methods (Rahmeyer, 2016; Vlados & Chatzinikolaou, 2020). Innovation is not born in a vacuum. Innovation always concerns the evolutionary matching between the production and demand potential (Figure 5).

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Figure 5. Innovation is the evolutionary outcome of matching supply and demand, based on Vlados and Chatzinikolaou
(Vlados & Chatzinikolaou, 2020).



Although it is primarily suggested that innovation comes as a market-pull or technology-push outcome, these two spheres combined give birth to the new condition and practice within any organizational system (Di Stefano et al., 2012; Peters et al., 2012). Specifically, supply refers to "what we can do," while demand to "what we want." Supply is about finding the proper resources and factors of production, synthesizing them, and offering products and services. On the contrary, demand refers to the unavoidable appearance of human needs that first transform to desires—and then to specific product demands.

Today's mega wave of innovation based on cyberphysical systems is associated with the continuous matching of demand for more sophisticated production systems with their respective supply in various socioeconomic organizations and industries. This perspective can be argued to be the core of the sought-after digital transformation in different industrial niches of the global economy (Hanelt et al., 2020). What is not clear is the context in which effective change management brought about by the diffusion of these new innovative labor and production systems can be achieved. In other words, the adoption of a new organizational adaptation framework is inevitable, something that the literature on the 4IR and the currently restructured labor relations does not seem to focus on, and this effort will be undertaken until the end of this section.

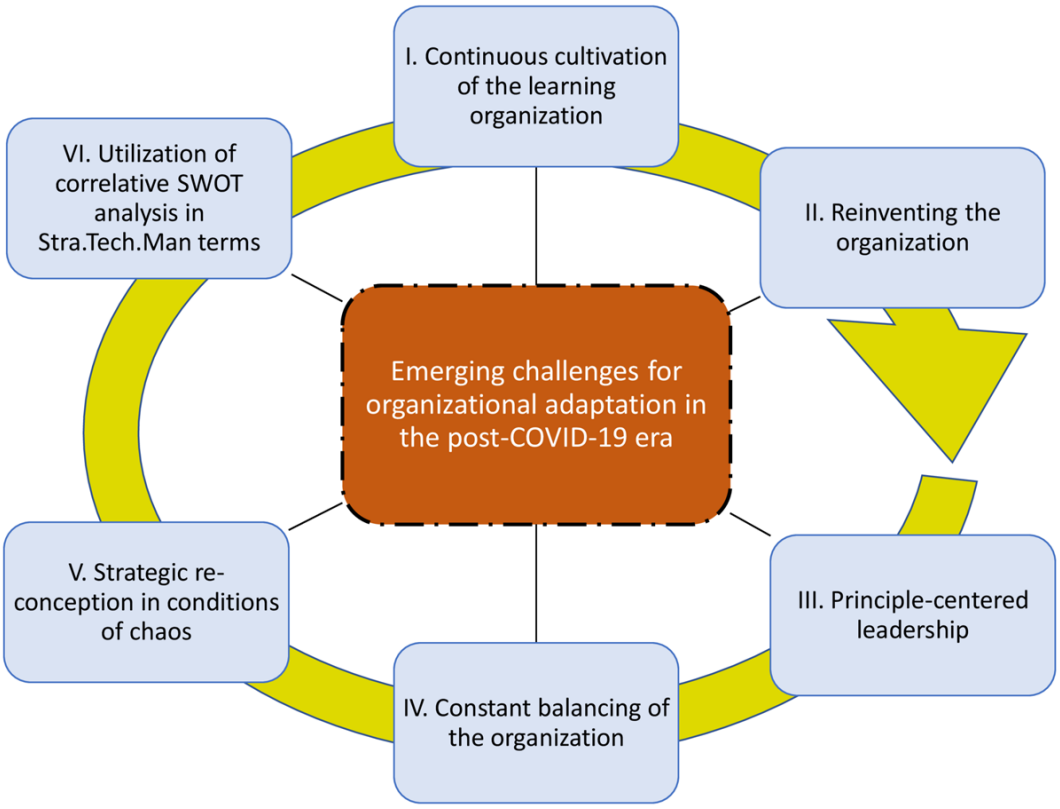
To this end, it must be noticed that socioeconomic organizations evolve and adapt—this realization has deep roots in the economic intellect. Already since Veblen (1898), it had been highlighted that a cumulative process of adaptation for all individuals in economic life exists, while Marshall (1890)—although the founder of neoclassical economics—had argued that "The Mecca of the economist lies in economic biology" (Hodgson, 1993). Concerning the organizational adaptation (OA) concept, the

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widely-cited work by Hrebiniak and Joyce (1985) is a significant point of reference. In this, it is pointed out that OA is about a multidisciplinary field because it "*is the subject of inquiry in organizational behavior, management, and economics*" (Hrebiniak & Joyce, 1985, pp. 346–347). Overall, it is suggested that evolutionary OA is about strategic choice and environmental determinism in the sense that "*adaptation is determined from without, as the environment selects organizations and allows only those forms with appropriate variations to remain*" (Hrebiniak & Joyce, 1985, p. 338). More recently, other scholars follow similar patterns, observing again the intentional decision-making involved, which leads to "*observable actions that aim to reduce the distance between an organisation and its economic and institutional environments"* (Sarta et al., 2021, p. 1).

This choice and environmental stimulus context are predominantly discussed as a new framework for OA amid the pandemic. This crisis offers opportunities for organizations to accelerate their transition and address various emerging challenges, including "*safe and secure digital information between informants and handlers, remote access to sensitive databases, electronic payment and transfer of informant-related finance, and enhancing the use of online communication (including recruitment and tasking)*" (Stanier & Nunan, 2021, p. 3). Therefore, the message of OA theorists is clear: the pandemic is a significant environmental change that requires the reintegration of organizational processes that pre-dated the outbreak (Khanal et al., 2021).

Figure 6. Emerging challenges for organizational adaptation in the post-COVID-19 era and a reconsideration of the remedies.
Authors' suggestion.



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Nevertheless, the context of OA for the post-pandemic era is still at the first stages of development in the literature. Therefore, significant contributions in organizational change management can illuminate the field of OA, aiming to offer remedies for treating "illnesses" that inevitably appear today in exogenously and endogenously transformed socioeconomic organizations (Vlados, 2019b). In general, today's need for managing change leads us to (re)update and (re)position some of the basic principles in the theory and practice of organizational adaptation for the post-COVID-19 era: (i) continuous fostering of the learning organization, (ii) systematic reinvention of the organizational perspective altogether, (iii) refocus on the principle-centered leadership, (iv) empowerment of continuous balancing of the organization, (v) re-conception of the strategy in conditions of chaos, and (vi) systematic utilization of correlative SWOT analysis in Stra.Tech.Man terms (Figure 6):

I. Continuous cultivation of the learning organization: This is the organization that all its members, irrespectively of hierarchy and expertise, are responsible for identifying, analyzing, and solving practical problems, allowing the organization to experiment, learn and continually improve. According to Senge (1990), the real challenge faced by humanity is our inability to grasp and manage human systems. The term "learning organization" refers to flexible, responsive, and adaptable organizations in their environment, which are not that bureaucratic. The fundamental element of this learning organization is the overwhelming increase in productivity that could manage to create and offer to its members a working environment that reflects their values. Senge stresses that hierarchy is still there even in extreme cases of highly networked organizations. However, this hierarchy is primarily of guiding principles and ideas, and, as a result, the hierarchical distribution of power in learning organizations is vastly different from traditional organizations. Overall, the term learning organization refers primarily to a step-by-step cognitive development for the entire organization.
II. Reinventing the organization: According to Goss et al. (1993), reinvention is a journey full of requirements and ups and downs. Those riding the "roller coaster" must be prepared for a ride full of challenges. When a company decides to follow the reinvention process, it must unveil and then change the hidden assumptions and foundations upon its decisions and actions are based because the unspoken past realities dictate what is available for the future. Therefore, for an organization to reinvent itself, it must uncover its "secrets." Organizations are usually failing because they base their future development only on their past practical experience. Goss et al. suggest that it is meaningless to do the same things repeatedly and expect different results—a problem found in complacent managers that tend to overconcentrate in the organization's past belief system. In the background, reinventing the organization means being capable of rigorous and patient self-criticism and not reacting like a threatened aristocracy that has to restore old "certainties."
III. Principle-centered leadership: According to Covey (1989), various defense mechanisms react in organizational transformation. When a company faces irreversible problems, all stakeholders must be expressing the firm's way of thinking. The goal is to understand what constitutes the company's consciousness because the principles and beliefs of managers about the company's strategy are less significant than the unconscious strategies implemented. From an overarching perspective, principle-centered leadership refers to realizing that leadership means the ability to embrace change and that changing requires unearthing and commitment to the organization's fundamental principles.
IV. Constant balancing of the organization: According to Duck (1993), the management model that is usually sufficient for day-to-day issues is not enough for change management since it resembles a patient that receives five surgical operations simultaneously, which can cause a death out of shock.

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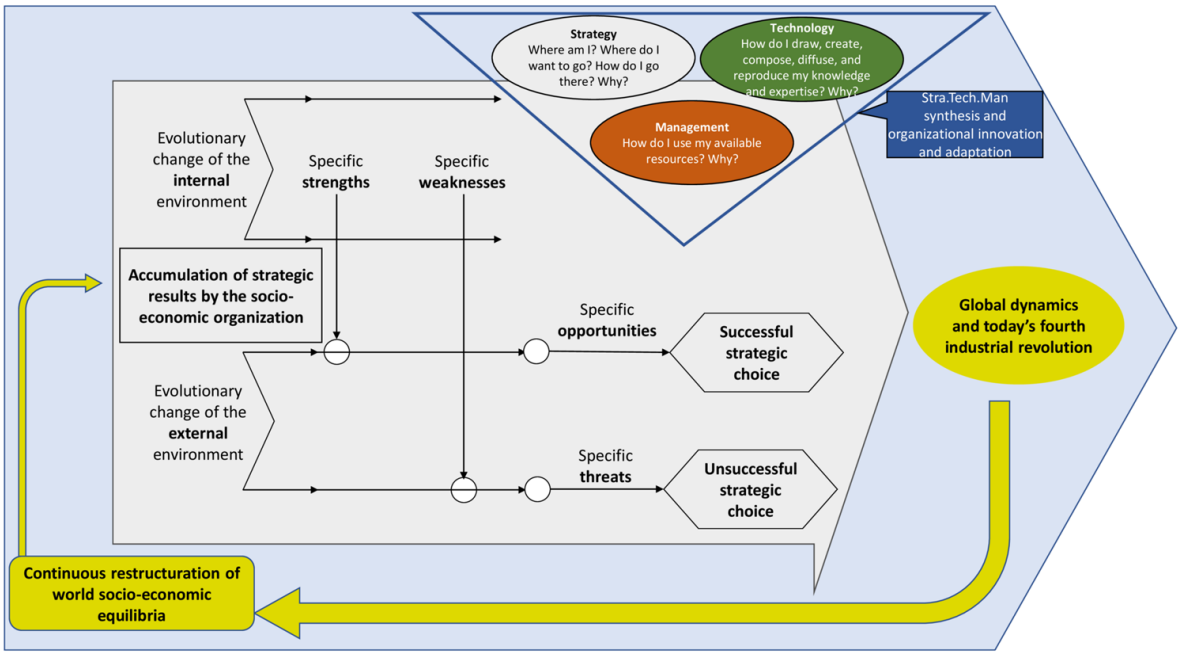
Breaking down change into smaller pieces is not enough since the management must handle the dynamics of change. Therefore, the real challenge is to bring innovations primarily in the intellectual sphere of work by simultaneously balancing all the pieces of the organizational puzzle. Most commonly, employees do not believe in the positive outcome of change and distrust the organization's new path. They tend to believe in organizational change only when they can see an action or behavior and the relevant results that confirm that the underlying change program is working. Overall, this "art of balancing" means that change management is an orchestrated effort that has to be undertaken by all stakeholders involved.

V. Strategic re-conception in conditions of chaos: According to Kotler and Caslione (2009), the world has entered an irreversible chaotic environment, and, therefore, most organization-wide improvement attempts have a significant impact on company performance. In the programs focusing on taking measures, a fundamentally flawed rationale tends to confuse the goals with the means and the process with the outcome. Various attempts to change end up with failure since the management might reject applicable improvement processes—by looking for the impossible—leaving finally empty-handed. The authors suggest setting up an integrated cycle of implementation and reaction in chaotic conditions and a plan for executing strategic behaviors to adapt to the new "chaotics." In generic terms, this chaotics approach emphasizes the continuous re-strategizing of the organization to gain an advantage among the competitors.

VI. Utilization of correlative SWOT analysis in Stra.Tech.Man terms: According to Vlados (2019a), to innovate, all organizations must exploit their comparative strengths, taking advantage of the opportunities presented to them over time. Accordingly, they must avoid the potential threats created by their "correlative" and evolutionary weaknesses. There are no shared opportunities and threats for all socioeconomic organizations—irrespectively of industrial/sectoral focus—and these must be understood and reexamined according to the historical formulation of the specific strengths and weaknesses. These accumulated strategic results lead to organizational adaptation combined with organizational technology and management (the "Stra.Tech.Man" synthesis). Moreover, this dynamic synthesis of organizational adaptation will determine whether various socioeconomic organizations will be left behind in development terms in the emerging idiosyncratic environment. All socioeconomic organizations only behaviourally can adapt because they are boundedly rational. Therefore, it is a fact that evolutionary biology has entered economic theorizing, and organizational innovation and adaptation for all socioeconomic organizations must be observed as the outcome of evolutionary synthesis between strategy, technology, and management. These correspond to the following sets of questions: Strategy is "*Where am I? Where do I want to go? How do I go there? Why?*" Technology is "*How do I draw, create, compose, diffuse, and reproduce my knowledge and expertise? Why?*" and Management is "*How do I use my available resources? Why?*" Finally, the correlative SWOT in terms of Stra.Tech.Man refers to finding the historically formulated and comparative strengths and weaknesses that lead to potential opportunities and threats based on clarifying the organization's strategic, technological, and managerial foundations (Figure 7).

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Figure 7. Organizational innovation and adaptation as the outcome of evolutionary synthesis between strategy, technology, and management. This figure extends the model presented in Vlados (2019a).



These are not the only approaches to change management, but they are prerequisites. They should be considered for improvement and organizational adaptation nowadays, even for small firms in the informal sector, who are the primary employers in many less-developed socioeconomic systems (Webb et al., 2020). An organization's choice depends on the strategy. It cannot be argued that a reinvention of the organization is "better" than the learning organization approach, nor that the chaos perspective is more effective than the Stra.Tech.Man theorization. However, it must be noticed that only the "Stra.Tech.Man approach" has been used as a diagnostic and consulting tool for smaller firms (Vlados & Chatzinikolaou, 2019). The other presented change management and strategy-rethinking perspectives concern primarily medium to larger organizations.

In conclusion, all socioeconomic organizations—irrespectively of their size—must readjust themselves nowadays. The fact that innovation must be the primary pursuit for the exit of the (continuous) crisis needs to be a common ground. However, innovation inescapably creates changes that must be managed in the most effective way possible.

CONCLUDING REMARKS

We authored this article both for the employees and the business practitioners that must adapt to the current pressing conditions created by the pandemic and the Fourth Industrial Revolution. Our goal was to highlight the changes occurring in the entire global socioeconomic system and how these affect the world of work and businesses. Based on the profound changes materializing nowadays in the labor environment, we showed that the pandemic accelerates the 4IR. The question that arose was whether organizational

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adaptation and change management could help the socioeconomic organizations (such as trade unions and employers) transform in the right direction. The current transition to the post-COVID-19 era shows that employees and employers must introduce change management and organizational adaptation methods.

The main suggestion by focusing on the micro-level was that the exit from the (continuous) crisis requires innovation, which unavoidably leads to changes that must be effectively managed. Based on this context, change management and OA approaches were examined as potential remedies for all socioeconomic organizations transformed today (employees, unions, and any other stakeholder of a production effort).

Overall, this study reaches specific conclusions that can prove significant towards understanding and promoting the required post-COVID-19 organizational adaptation. The following notes constitute the epicenter of this chapter's contribution:

1. The COVID-19 crisis causes significant transformations in the global socioeconomic system. According to recent data, the pandemic crisis and the entailing socioeconomic crisis have driven an unprecedented recession and structural destabilization of various institutions. As it seems, the expected recovery will not lead in the foreseeable future to achieve the envisaged pre-pandemic growth rates globally. As a result, the likely growth outcome seems of "L-type"—most indices and perspectives for the less developed and resilient socioeconomic systems will move at comparatively lower levels, ending with relatively weaker growth and development rates in the immediate aftermath of this crisis.
2. The acceleration of the 4IR leads irreversibly nowadays to the restructuration of the working environment, generating notable challenges on the future perception of human resources and labor (immediate and far future). The most significant readjustment concerns the continuous digital transformation that seems imperative to all socioeconomic systems, with the diffusion of the "cyberphysical systems" leading the less adaptive employees and firms to the need for building new skills and knowledge.
3. The exit of this socioeconomic crisis (after the direct consequences of the pandemic) seems to require the various actors to reposition their organizational adaptation ability and resilience, emphasizing the strategic structuration of new digital skills and the need for constant innovation in generic and holistic terms. In this context of the necessitated innovational reinforcement, the generation of the prerequisite change management mechanisms is vital to all socioeconomic organizations and systems (businesses, workers, and institutions globally).

These points lead to perceive the necessary effort for fertile adaptation into the new conditions in simultaneous micro, meso, and macro contexts. The partial and fragmentary perspectives are relatively insufficient to offer a sustainable future solution. As it seems, systematic training programs for employees are a significant policy direction for the future, supporting the various business ecosystems to reinforce their adaptation and resilience potential (Vlados, 2021). However, this kind of approach to either international strategies or various national policies to strengthen the different local and industrial systems that host entrepreneurial activity falls into the scope of a future study.

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KEY TERMS AND DEFINITIONS

Evolutionary Organizational Adaptation: Even though adaptation is determined externally since the environment selects organizations and allows appropriate variations to survive, organizational adaptation derives from environmental determinism and strategic choice simultaneously.

Fourth Industrial Revolution: This new reality builds upon the third industrial revolution, driving the digital revolution that started in the late twentieth century and the further socioeconomic deployment of the "cyber-physical systems," increasingly blurring the boundaries between the physical, digital, and biological spheres.

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Neo-Schumpeterian Innovation: It concerns the evolutionary matching between the production and demand potential, causing the "creative destruction" of older and aging organizational configurations due to the "entrepreneur's ambitions" who, in search of profit, is driven to the introduction of new products and new production and marketing methods.

Post-COVID-19 Change Management: It concerns the constant (re)update and (re)position of the fundamental change management principles, such as continuous fostering of the learning organization, reinventing the organizational perspective, refocusing the principle-centered leadership, reinforcing the organizational balance, reconceptualizing the strategy in conditions of chaos, and utilizing the correlative SWOT analysis in Stra.Tech.Man terms (strategy-technology-management synthesis).

Post-COVID-19 Era: This evolutionary phase concerns the period emerging after the COVID-19 crisis, which leads the global socioeconomic system and the various actors at all levels (macro-meso-micro), respectively, to irreversible transformations, laying the ground for the appearance of an entirely new form of globalization.

Post-COVID-19 Working Environment: This new post-COVID-19 framework signals a new labor market characterized by significant divergence regarding wage and educational level primarily in tele-workability, requiring now the continuous advancement of digital competencies for future workplaces.

ENDNOTES

1 However, extremism in politics is not new: See the research by Van Der Brug and Fennema (2003) and Arzheimer and Carter (2006).

2 And other approaches to global transformation have also been developed besides the 4IR, although not as popular, even though they have significant interpretive power. The framework of the "new globalization" is one relevant perspective (Bhattacharya et al., 2017; Vlados, 2019c).

3 See the related concept of "prosumer" for the future energy sector: Kotilainen et al., 2016.